Max Coursey

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SUMMARY

Senior AI/ML leader currently architecting enterprise-grade transformation at a major energy infrastructure company, delivering multi-million dollars in operational savings through multi-agent systems, MLOps automation, and intelligent optimization platforms. Combines deep technical execution (building production ML systems from zero) with proven executive experience managing distributed teams, vendor relationships, and regulatory compliance including SOX implementation. Uniquely positioned to scale AI initiatives across industrial operations while maintaining enterprise governance and security standards

EXPERIENCE

Senior ML/Data Engineer & AI Solutions Architect | Colonial Pipeline | Atlanta, Ga | April 2025 - Present

- Developed a **natural language query interface** enabling non-technical pipeline operators to request sensor data through conversational prompts without SQL knowledge, prototyping in Vertex AI Workbench with Gemini models before transitioning to deterministic rule-based prompting for reliable BigQuery SQL generation, deploying a production interface as a **containerized TypeScript API on Cloud Run** integrated with a web application frontend
- Implemented **automated data validation workflows** with configurable business rules enforcing operational limits across pipeline segments, designing a centralized business rules table in BigQuery with temporal versioning, maintaining a complete audit history of threshold changes, driving Dataflow streaming pipelines that dynamically route sensor data through transformation logic and ML prediction endpoints, with analytics queries leveraging **temporal joins** to assess how threshold adjustments impact detection accuracy over time
- Developed **machine learning models for pump anomaly detection,** progressing from BigQuery ML statistical methods (78% accuracy) to Vertex AI AutoML Tables (85% accuracy), currently evaluating production ensemble architecture combining Isolation Forest, LSTM neural networks, and XGBoost with SHAP-based explainability to identify causal sensor combinations driving predictions
- Engineered **real-time ML inference pipeline** using Dataflow streaming jobs processing IoT sensor data with automated feature engineering (rolling statistics, rate-of-change calculations, cross-sensor correlations) before routing to Vertex AI prediction endpoints
- Enabled advanced analytics across **industrial IoT infrastructure** by architecting an intelligent sensor data transformation system that converts sparse, exception-driven streams into continuous time-series datasets using statistical ML models, automated outlier detection, and business-context-aware flagging algorithms
- Delivered and managed **enterprise AI transformation** across multiple business units, architecting solutions that generated **$1M+ in annual operational savings** through automated vendor selection, optimized fuel blending, and intelligent procurement processes while maintaining strict regulatory compliance
- Orchestrated **MLOps pipelines on Vertex AI** with automated CI/CD workflows, custom containerization, and secure deployment practices for regulated industrial environments, enabling scalable ML model lifecycle management

ML/Senior Data Engineer | Gordon Food Services | Advanced Analytics | 2022 - April 2025

- **Transformed an experimental TensorFlow notebook into an enterprise-grade forecasting system**, architecting robust MLOps infrastructure on Vertex AI with automated training pipelines, GPU acceleration, and distributed processing for scalable deployment.
- **Optimized model architecture** using advanced programming and constraint optimization, implementing concurrent and mixed-integer programming, early stopping, and gradient descent to achieve a **4% performance improvement over O9 Solutions' baseline model**.
- **Deployed a production forecasting solution projected to deliver $1M in annual savings**, validated through comprehensive **five-year historical backtesting** and rigorous performance analysis.
- **Implemented automated model performance monitoring tools**, enabling Data Scientists to leverage Vertex AI Experiments with TensorBoard for demand forecasting and Vizier for streamlined hyperparameter tuning workflows.
- **Engineered end-to-end ML pipelines on GCP**, utilizing Vertex AI for time series forecasting with custom TensorFlow, PyTorch, Prophet, and GreyKite models while supporting automated deployment capabilities.
- **Built comprehensive ML infrastructure** including feature selection tools, distributed model training, and support for custom models, BigQuery ML, and AutoML, enabling streamlined deployment and **real-time inference**.
- **Enhanced model training efficiency by 25%** through optimized data access and storage, creating feature stores on BigQuery and leveraging managed datasets and metadata frameworks in GCP.
- **Reduced BigQuery costs by 10% monthly** through strategic partitioning, clustering, and slot usage analysis on core data science tables, demonstrating advanced SQL optimization and cost management expertise.
- **Implemented a comprehensive data governance strategy** using GCP Dataplex with hierarchical metadata tagging and labeling, establishing **data mesh architecture** that enables domain-driven data ownership and discovery across industrial operations.

Research Engineer/Developer | Vanderbilt University | ScopeLab AI Research Lab | 2021 - 2022

- **Architected scalable ETL pipeline infrastructure** using PySpark frameworks, processing **TB+ of diverse academic and transportation datasets** across 5+ concurrent research projects while establishing automated data preparation workflows that **reduced research scientist setup time from weeks to hours**.
- **Built an enterprise-grade data standardization and ingestion platform** leveraging PySpark SQL and Pandas, creating reusable pipeline templates that **accelerated retraining cycles by 60%** and enabled 5+ researchers to rapidly prototype and deploy solutions.
- **Established comprehensive cloud infrastructure** supporting 5+ research scientists, managing AWS/GCP environments including Jupyter notebooks, compute instances, and distributed processing pipelines with **90%+ uptime SLA** for scalable machine learning experimentation.
- **Developed a full-stack transportation optimization platform** with a **real-time vehicle tracking system** and ML-based route optimization, serving 50+ daily shuttle operations while integrating manual routing adjustments and predictive analytics that **improved operational efficiency by 25%** and reduced passenger wait times.

Technical Lead & Founding Engineer | Mobius.ai | (Transportation optimization SaaS) | 2019 - 2021

- **Sole technical architect** responsible for end-to-end platform development, from initial architecture through production deployment.
- **Built complete data infrastructure** processing **TB+ datasets**, an ML operations platform, and full-stack applications serving 50+ daily users.
- **Owned the entire technical lifecycle**: architecture design, implementation, model integration, user testing, and iterative improvements based on customer feedback.
- **Established scalable foundations** that enabled 5+ researchers to rapidly prototype and deploy ML solutions.
- The company remains profitable and has operated successfully for over five years.

Director, Business Technology Solutions | Grant Thornton | 2014 – 2019

- **Led a team of 5-10 engineers and consultants**, managing a **$1-1.5M annual project portfolio** across large corporate clients.
- **Delivered $50M+ in client value** through digital transformation initiatives surrounding tax and financial strategies with large-scale commercial real estate, technology companies, and automated outsourcing solutions.
- **Drove digital transformation initiatives** that generated **multi-million-dollar client savings** by automating tax/financial calculations, aggregating data, and optimizing tax positioning strategies.
- **Established technical architecture and governance frameworks** to ensure GAAP/IFRS compliance and regulatory adherence, providing domain expertise and maintaining strict data integrity and audit trail requirements across financial systems.
- **Built enduring client relationships** from startups to large corporations, translating technical capabilities into business value and maintaining client communication throughout project lifecycle.

Manager, National Leader Tax Technology Solutions | Cherry Bekaert, LLP | 2012 – 2014

- **Supported practice growth by architecting client acquisition processes**, contributing to doubling practice size and **generating $2M+ annual revenue** across 30+ concurrent engagements.
- Managed geographically diverse team, processes, and sales pipeline for tax fixed asset and R&D tax credit projects.
- Designed and implemented new statistical sampling process for repairs and R&D credit projects reducing project labor expenses by 30-60%.
- Managed all project aspects including initial scoping, budgeting, project planning and control, delivery, internal software/tool development, and internal training.

Senior Associate, Tax Technology & Integration Services | EY, LLP | 2007 - 2011

- **Architected an automated SQL-based fixed asset workflow solution** for the **$12B MGM CityCenter mega-project**, producing 3x results over peers.
- Developed historical client/project database with simple linear regression model to accurately estimate potential benefits and identify industries, clients, and project types delivering the highest margin.
- Consistently achieved **top-tier performance recognition** (5/5 ratings, top 15% firmwide), demonstrating technical leadership excellence and strategic value delivery in high-stakes enterprise technology implementations.

EDUCATION

Master of Science, Computer Science | Vanderbilt University

Master of Business Administration | Georgia State University

Bachelor of Science, Environmental Engineering | Georgia Institute of Technology

Bootcamp | MongoDB, Express, React, Node.js | Georgia Institute of Technology

CERTIFICATIONS

AWS Certified Developer Associate | 2019-2021

SKILLS

ML/AI:
Python, TensorFlow, BigQuery ML, Vertex AI, Ensemble Methods, Feature Engineering, Time-Series Analysis, Anomaly Detection, Generative AI

GenAI & LLM:
Gemini API, Prompt Engineering, Rule-based Agents, Natural Language to SQL, RAG Architectures, LangChain, Agent Orchestration, Function/Tool Orchestration

Data Engineering:
BigQuery, SQL, SQLx, Apache Beam, Feature Stores, Data Modeling, Data Validation, Golden Tables, Data Governance

Cloud & MLOps:
GCP (Vertex AI, Cloud Run, BigQuery, Dataflow), MLOps CI/CD, Docker, Automated Retraining, Model Monitoring, Drift Detection, Production ML Deployment & Ongoing Evaluation

Development:
TypeScript, JavaScript, Python (FastAPI, Flask), React, Node.js, RESTful APIs, Full-stack Development, Dash Plotly

I build high-performing teams through direct communication, pragmatic problem-solving, and creating environments where people feel genuinely valued and can bring their authentic selves to solve complex technical challenges.